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PROGRAM 2: FINITE STATE AUTOMATON

AIM

To implement a finite state automaton (FSA) that accepts strings ending with 'ab'.

ALGORITHM

• Start the program.

• Read a string from the user.

• Initialize the current state as 0.

• For each character, update the state according to the transition rules.

• If the final state is 2, display “String Accepted”; otherwise, display “String Rejected”.

• Stop the program.

PSEUDOCODE

```
START
READ text
state = 0
FOR each character ch in text
    IF state = 0
        IF ch = 'a', state = 1
        ELSE, state = 0
    ELSE IF state = 1
        IF ch = 'a', state = 1
        ELSE IF ch = 'b', state = 2
        ELSE, state = 0
    ELSE IF state = 2
        IF ch = 'a', state = 1
        ELSE, state = 0
IF state = 2
    PRINT "String Accepted"
ELSE
    PRINT "String Rejected"
STOP
```

PROGRAM

```
text = input("Enter a string: ")

state = 0

for ch in text:
    if state == 0:
        if ch == 'a':
            state = 1
        else:
            state = 0

    elif state == 1:
        if ch == 'a':
            state = 1
        elif ch == 'b':
            state = 2
        else:
            state = 0

    elif state == 2:
        if ch == 'a':
            state = 1
        else:
            state = 0

if state == 2:
    print("String Accepted")
else:
    print("String Rejected")
```

OUTPUT

Output screenshot:

```
text = input("Enter a string: ")

state = 0

for ch in text:
    if state == 0:
        if ch == 'a':
            state = 1
        else:
            state = 0

    elif state == 1:
        if ch == 'a':
            state = 1
        elif ch == 'b':
            state = 2
        else:
            state = 0

    elif state == 2:
        if ch == 'a':
            state = 1
        else:
            state = 0

if state == 2:
    print("String Accepted")
else:
    print("String Rejected")
```

```
Enter a string: abab
String Accepted
```

RESULT

Thus, the finite state automaton was implemented successfully to accept strings ending with 'ab'.